QUALIFICATION STANDARD AND PLAN

Naval Remote Weapon System Operator

Boatswain 00105

MITE ID – 121750

RECORD OF AMENDMENTS

1. The Training Authority (TA) designated for this qualification is the Commander Naval Personnel and Training Group (CNPTG). Naval Training Development Centre Atlantic (NTDC(A)) is the designated development centre responsible for the maintenance of training. Amendments to Performance Objectives (POs) are to be requested through NTDC(A) and approved by CNPTG. Amendments to Lesson Guidance (LG) are to be approved by the Commanding Officer (CO) of NTDC(A).

2. Qualification Standard and Plan (QSP) versions with a 1 preceding the decimal indicate legacy 4 Chapter documents. QSP versions with a 2 preceding the decimal indicate Firefly 2 Chapter documents. QSP versions with a 3 preceding the decimal indicate 3 Chapter documents. Versions with a number followed by a bracketed number indicate a stray from this nomenclature. The number in brackets indicates the corrected version number.

Version Number	Amendment Type	Approval Date	Authority
1.00.00	Initial	18 Nov 2022	CNPTG

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ACRONYMS AND ABBREVIATIONS

A

ATT	Actual Training Time

B

BAT	Base Administrative Time
BIT	Built in Test

C

CAF	Canadian Armed Forces
CCD	Charged Coupled Display (Day Camera)
CBT	Computer Based Trainer
CFITES	Canadian Forces Individual Training and Education System
Cmdt	Commandant
CNPTG	Commander Naval Personnel and Training Group
CO	Commanding Officer
CSNI	Consolidated Secret Network Infrastructure

D

DAOD	Defence Administrative Orders and Directives
DSCDS	Defence Subject Classification and Disposition System
DND	Department of National Defence
DPGR	Director Personnel Generation Requirements
DWAN	Defence Wide Area Network
DU	Display Unit

E

EC	Enabling Check
EDI	Explain, Demonstrate, Imitate
EL	Elevation
EOSS	Electro Optic Sensor Suite
EO	Enabling Objective

F

FEK	Firing Enabling Key
FLIR	Forward-Looking Infra-Red (Night Camera)
FW	Forward

G

GPNet	General Purpose Network

H

HOD	Head of Department
HMG	Heavy Machine Gun

I

IAW	In Accordance With
ID	Identification
IMS	Information Management Standard
IS	Information Systems
IT	Information Technology

L

LO	Learning Objective
LG	Lesson Guidance
LRF	Laser Range Finder
M	
MITE	Military Individual Training and Education
MOSID	Military Occupational Structure Identification Code
MCU	Main Control Unit
MCP	Main Control Panel
MO	Manual Override
M2HB	Mark 2 Heavy Barrel
N	
NQual	National Qualification
NFS	Naval Fleet School
NAVORD	Naval Order
NPTG	Naval Personnel and Training Group
NPTORD	Naval Personnel and Training Order
NTDC(A)	Naval Training Development Centre (Atlantic)
N/A	Not Applicable
NRWS	Naval Remote Weapon System
O	
OJT	On-the-Job Training
OJPR	On-the-Job Performance Requirement
P	
PC	Performance Check
PO	Performance Objective
PLAR	Prior Learning Assessment and Recognition
PT	Physical Training
Q	
QSP	Qualification Standard and Plan
QCB	Quick Change Barrel
R	
R&Q	Rations & Quarters
RCN	Royal Canadian Navy
ROC	Remote Operator Console
S	
SAT	Satisfactory
SO	Standing Order
SOP	Standard Operating Procedure
T	
TAT	Training Administrative Time
TA	Training Authority
TD	Temporary Duty
TE	Training Establishment
TEL	Technology Enabled Learning

TO	Training Officer
TPO	Training Petty Officer
TRB	Training Review Board
TS	Training Site
TTT	Training Travel Time
U	
UNSAT	Unsatisfactory
V	
Vol	Volume
W	
WS	Weapon Station
WSS	Weapon System Simulator

FOREWORD

3. This training is based on the specification found at Director Personnel Generation Requirements (DPGR) website: http://cmp-cpm.mil.ca/en/support/military-personnel/dpgr-index.page.

4. There is no National Qualification (NQual) associated with this training.

5. A trainee will be granted the Military Individual Training and Education (MITE) identification (ID) 121750 when standards listed in all POs have been met.

6. The designated TEs for this training include all NFSs within NPTG.

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CHAPTER 1: TRAINING MANAGEMENT DETAILS

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AIM

1. The aim of this QSP is to establish the requirements and training for the Naval Remote Weapon System Operator (MITE: 121750). This QSP is designed to assist NTDC(A) in developing, NFSs in conducting and evaluating, and NPTG in validating the training.

COURSE DURATION

2. Course duration is calculated in accordance with (IAW) Naval Personnel and Training Order (NPTORD) 4500-22 Annex B Table 3.1. Partial days are rounded up. For this course, a period of instruction is 50 minutes in duration and each day has 8 instructional periods. The course duration calculation is based on maximum load.

Course Duration Calculation		
Allocation of Actual Training Time (ATT)	Periods	Days
Lesson Guidance (LG)	30	3.75
Enabling Checks (ECs)	10	1.25
Performance Checks (PCs)	12	1.5
ATT SUB TOTAL:	**52**	**6.5**
Allocation of BAT, TAT, TTT and PT	Periods	Days
Base Administrative Time (BAT)	1	
Training Administrative Time (TAT)	1	
Training Travel Time (TTT)	0	
Physical Fitness (PT)	0	
BAT, TAT, TTT, PT SUB TOTAL:	**2**	**.25**
***Partial days are rounded up** **TOTAL:**	**54**	**7**

REQUIRED COURSE STAFFING

3. The following personnel, including guest lecturers, are required to conduct this training:

Position	Rank	Quantity	Qualification(s)	Security Requirement(s)	PO/EO/LG
Course Instructor	MS-PO2	1	RQ-PO2 BOSN NRWS Operator	SEC II	All
Course Staff	S1-MS	1	RQ-S1 BOSN PLQ NRWS Operator	SEC II	All
Guest Lecturer	S1-MS	1	RQ-S1 CSE NRWS Maintainer MITE: 121749	SEC II	EO 001.02

COURSE CAPACITY

4. Planning for the conduct of training is based on availability of resources; course capacities are as follows:

5. Maximum – 8;

6. Minimum – 2; and

7. Optimum – 8.

PREREQUISITES

8. To be selected for this course, the loading authority must ensure that trainees have the following prerequisites:

9. BOSN RQ-S3 (ALTO or equivalent);

10. .50 Cal HMG M2HB QCB Operator (MITE: 118113); and

11. C6A1 GPMG Operator (MITE: 118108).

PRIOR LEARNING ASSESSMENT AND RECOGNITION

12. Prior Learning Assessment and Recognition (PLARs) shall be conducted IAW Naval Order (NAVORD) 4500-2, Defence Administrative Orders and Directives (DAOD) 5031-1, and Canadian Forces Individual Training and Education System (CFITES) Volume (Vol) 12.

SPECIAL INSTRUCTIONS

13. Nil.

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PO 001 – ENABLE SYSTEM FOR WEAPON USE

1. **Performance:** Enable System For Weapon Use.

2. **Conditions:**

   a. Given:

      (1) Cue: Any necessary reason to take and pass control of the mount;
      (2) NRWS Mount;
      (3) .50 Cal HMG M2HB QCB;
      (4) C6A1 GPMG;
      (5) NRWS weapon replacement kit;
      (6) Links; and
      (7) Linker/delinker.

   b. Denied: Supervision; and

   c. Environment:

      (1) All weather conditions;
      (2) At sea or alongside;
      (3) Domestic or deployed ops; and
      (4) Day or night.

3. **Standard:** The NRWS Operator shall Enable System For Weapon Use IAW A1, A2, A3, A4, A5, A7, A9, A10, A11, A12 by:

   a. Mounting .50 Cal HMG M2HB QCB/C6A1 GMPG by:

      (1) Identifying major components of the mount;
      (2) Ensuring manual override is set to safe;
      (3) Placing weapon on mount;
      (4) Placing retaining pins in correct location based on weapon; and
      (5) Securing retaining pins.

   b. Converting the NRWS mount between manual and remote from the stowed position by:

      (1) Ensuring manual override set to safe;
      (2) Engaging mount servos;
      (3) Disengaging locking pins; and
      (4) Disengaging BLR and weapon safety.

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4. Training Limitations: Nil.

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EO 001.01 – MOUNT MACHINE GUN ON WEAPON STATION
LG 001.01.01 – IDENTIFY MAJOR COMPONENTS

LG 001.01.01 – Identify Major Components		
Location Conditions	**Instructional Sequence**	
Dedicated NRWS Classroom	Beginning of course	
Time	**Classification**	**Instructor**
50 minutes	UNCLASS	RQ PO2 BOSN RQ S1 BOSN
Learning Objectives	**Instructional Method(s)**	
1) Identify major components of the mount	1) EDI	
Scenario	A new junior member has been posted in to your Deck Department. Your task is to provide them with an overview of the NRWS mount and how to configure it in advance of proceeding to sea.	
Activity Steps	1) Introduction (5 minutes) Instructor shall introduce the NRWS to trainees including system's intended use in force protection, surveillance, SAR, MOB, etc. 2) Performance (35 minutes - EDI) Instructor shall gather trainees at the NRWS mount to identify and explain the following major components of the NRWS mount: 1. **Manual override (MO) switch (CRITICAL FACTOR);** 2. Weapon Station (WS) cover; 3. Charging mechanism motor; 4. Elevation actuator; 5. Dual ammunition box; 6. Cradle base; 7. Weapon cradle components; 8. Washer fluid reservoir; 9. Washing nozzle; 10. EOSS mount; 11. EOSS; 12. Service door; 13. Weapon replacement kit; 14. Locking pins; 15. Hard stops; 16. Handles for manual firing; and 17. Fire triggers. 3) Conclusion (10 minutes) Instructor will have trainees come up individually to point out and explain one major component of the mount until all material is covered.	

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LG 001.01.01 – Identify Major Components	
Instructor Resources	**References**
1) Fully equipped NRWS mount	1) A1, A2, A3, A4, A5, A7, A9, A10, A11, A12
Note(s) to Instructor	
1) Instructor will emphasize critical safety importance of the manual override switch.	

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LG 001.02.01 – MOUNT MACHINE GUN ON WEAPON STATION

LG 001.01.02 – Mount Machine Gun on Weapon Station		
Location Conditions	**Instructional Sequence**	
Dedicated NRWS Classroom	After LG 001.01.01	
Time	**Classification**	**Instructor**
150 minutes	UNCLASS	RQ PO2 BOSN RQ S1 BOSN RQ S1 CSE
Learning Objectives	**Instructional Method(s)**	
1) Mount .50 Calibre (Cal) Heavy Machine Gun (HMG) Mark 2 Heavy Barrel (M2HB) Quick Change Barrel (QCB) and C6A1 General Purpose Machine Gun (GPMG)	1) EDI	
Scenario	HMCS CHATHAM is proceeding to sea and you are tasked to Mount Machine Gun on Weapon Station for pre sail checks.	
Activity Steps	1) Introduction (5 minutes) 1. Instructor will explain that the NRWS can be set up for use with either .50 Cal HMG M2HB QCB or C6A1 GPMG; explain that as an operator, trainees will be required to know how to install both types of weapons on the weapon cradle. 2. Instructor will explain that the mount requires a weapon replacement kit to go between .50 Cal HMG M2HB QCB and C6A1 GPMG. Bosns will assist CSE in the replacement process. 2) Performance/Activity (5 minutes – EDI) Instructor shall explain and demonstrate the mounting of the .50 Cal HMG M2HB QCB by: 1. Ensuring MO set to safe; 2. Open WS cover; 3. Lowering Firing Mechanism; 4. Placing weapon on mount; 5. Placing retaining pins in correct location based on weapon; 6. Securing retaining pins; 7. Raise and pin Firing Mechanism; and 8. Close WS Cover. 3) Performance/Activity (40 minutes – EDI) Instructor will remove weapon and assist Guest Speaker in conducting replacement to C6A1 GPMG and back to .50 Cal HMG M2HB QCB by: 1. Utilizing weapon replacement kit and replacement parts contained within; 2. Placing C6A1 GPMG on mount;		

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LG 001.01.02 – Mount Machine Gun on Weapon Station	
	3. Placing retaining pins on correct location based on weapon; 4. Securing retaining pins; 5. Utilizing weapon replacement kit to return to .50 Cal HMG M2HB QCB state; and 6. MO to live 4) Performance/Activity (95 minutes – EDI) Instructor shall have trainees in pairs, mount the .50 Cal HMG M2HB QCB, and identify parts within the kit. 5) Conclusion (5 minutes) Instructor shall summarize the LG, resolve any issues encountered during the lecture and ask confirmation questions of the trainees.
Instructor Resources	**References**
1) NRWS Mount 2) .50 Cal HMG M2HB QCB 3) C6A1 GPMG	1) A1, A2, A3, A4, A5, A7, A9, A10, A11, A12
Note(s) to Instructor	
1) Guest Speaker shall be CSE NRWS Maintainer. (MITE: 121749) 2) Instructor to ensure emphasis on the manual override is set to safe when approaching mount. (**CRITICAL FACTOR**) 3) Guest Speaker to be arranged ahead of time. Guest Speaker to lead discussion, demonstration, and supervise trainees in conversion of mount to and from C6A1 GPMG and .50 Cal HMG M2HB QCB. 4) Instructor to ensure all necessary tools and equipment are prepared and on site prior to commencing this EO. 5) Upon completion of LG, mount to be set up to accept the .50 Cal HMG M2HB QCB.	

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EO 001.02 – TRANSFER CONTROL OF THE WEAPON STATION
LG 001.02.01 – TRANSFER CONTROL OF THE WEAPON STATION

LG 001.02.01 – Transfer Control of the Weapon Station		
Location Conditions	**Instructional Sequence**	
NRWS Classroom	After LG 001.02.01	
Time	**Classification**	**Instructor**
100 minutes	UNCLASS	RQ PO2 BOSN RQ S1 BOSN
Learning Objectives	**Instructional Method(s)**	
1) Transfer mount between manual and remote operation	1) EDI	
Scenario	HMCS NISGA'A is proceeding to sea and you are tasked to confirm manual operation of the NRWS mount for pre sail checks.	
Activity Steps	1) Introduction (5 minutes) Instructor will explain the steps required to transfer the mount between manual and remote operation to include: 1. Ensuring manual override set to safe (**CRITICAL FACTOR**); 2. Engaging/Disengaging mount servos; 3. Engaging/Disengaging locking pins; 4. Engaging/Disengaging BLR and weapon safety; and 5. Discussion on drills utilized from NCPM 704 when in manual operation. 2) Performance/Activity (90 minutes – EDI) 1. Instructor will demonstrate how to transfer the mount between manual / remote, and conducting drills with the .50 Cal HMG M2HB QCB. 2. Individually, trainees will transfer the mount from remote to manual, conduct drills, and then return the mount to remote operation. 3) Conclusion (5 minutes) Instructor to summarize LG and reinforce important critical factors.	
Instructor Resources	**References**	
1) NRWS Mount 2) .50 Cal HMG M2HB QCB 3) Practice rounds 4) Links 5) Linker/delinker	1) A1, A2, A3, A4, A5, A7, A9, A10, A11, A12	
Note(s) to Instructor		
1) Emphasise that manual override is set to safe when approaching mount.(**CRITICAL FACTOR**) 2) If time permits, conduct NCPM 704 drills to give manual use experience.		

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EC 001 – ENABLE SYSTEM FOR WEAPON USE

EC 001 – Enable System For Weapon Use		
Scope	**Location Conditions**	**Instructional Sequence**
EO 001.01-EO001.03	Dedicated NRWS Classroom	After LG 001.03.01
Time	**Classification**	**Instructor**
100 minutes	UNCLASS	RQ PO2 BOSN RQ S1 BOSN
Scenario	HMCS ESSEX is preparing to set sail. As part of pre-sail checks, you are tasked to ensure the NRWS mount operates in manual mode.	
Assessment Steps	**Input** – Fully equipped NRWS mount, NRWS weapon replacement kit. **Performance** – IAW EC checklist. (12 minutes) **Output** – Trainee must achieve 11/14 SATs INCLUDING ALL CRITICAL FACTORS to pass this assessment. **Feedback** – Group feedback upon completion, IAW EC checklist.	
Instructor Resources		
1) Fully equipped NRWS mount. 2) NRWS weapon replacement kit.		
Note(s) to Instructor		
1) Any remaining time to be spent in feedback. 2) Must review Chapter 3, Enabling and Performance Checks and their definitions located in NPTORD 4500-54 prior to assessment of trainee.		
Assessment Limitations	Nil.	
Impact of Failure	IAW NPTORD 4500-54.	

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PROTECTED B (when complete)

EC 001 ENABLE SYSTEM FOR WEAPON USE			
Item	**SAT**	**UNSAT**	**Remarks**
MOUNT .50 CAL HMG M2HB QCB			
Manual override set to safe	☐	☐	**CRITICAL**
Place weapon on mount	☐	☐	
Place retaining pins in correct location	☐	☐	
Secure retaining pins	☐	☐	
CONVERT THE NRWS MOUNT FROM MANUAL TO REMOTE			
Manual override set to safe	☐	☐	**CRITICAL**
Disengage mount servos	☐	☐	
Disengage locking pins	☐	☐	
Disengage BLR	☐	☐	
Engage weapon safety	☐	☐	
CONVERT THE NRWS MOUNT FROM REMOTE TO MANUAL			
Disengage weapon safety	☐	☐	
Disengage BLR	☐	☐	
Engage mount servos	☐	☐	
Disengage locking pins	☐	☐	
Set MO to live	☐	☐	**CRITICAL**

Score: /14

Trainee must achieve 11/14 SATs INCLUDING ALL CRITICAL FACTORS to pass this assessment.

The trainee must achieve x/y SATs to pass this assessment.

Positive Feedback	**Improvement Feedback**

Overall Assessment: (Circle One)	**Met Standard**	**Did Not Meet Standard**

Assessor
I have discussed the results of this assessment with the trainee.

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EC 001 ENABLE SYSTEM FOR WEAPON USE		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

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PO 001 – ENABLE SYSTEM FOR WEAPON USE
PC 001 – ENABLE SYSTEM FOR WEAPON USE

PC 001 – Enable System For Weapon Use		
Scope	**Location Conditions**	**Instructional Sequence**
PO 001	Dedicated NRWS Classroom	After EC 001
Time	**Classification**	**Instructor**
150 minutes	UNCLASS	RQ PO2 BOSN RQ S1 BOSN
Scenario	A new junior member has been posted in to your Deck Department. Your task is to provide them with an overview of the NRWS mount and how to configure it in advance of proceeding to sea.	
Assessment Steps	**Input** – Fully equipped NRWS mount, .50 Cal HMG M2HB QCB, NRWS weapon replacement kit. **Performance** – IAW PC checklist. (13 minutes) **Output** – Trainee must attain 11/14 SATs INCLUDING ALL CRITICAL FACTORS to pass this assessment. **Feedback** – Immediate feedback upon completion (2 minutes)	
Instructor Resources		
1) Fully functional NRWS trainer 2) NRWS weapon replacement kit 3) .50 Cal HMG M2HB QCB		
Note(s) to Instructor		
1) Remind students to make the verbal declaration noted in test. 2) Although not tested, trainees must verbally name each component they interact with IOT improve their nomenclature of mount components. 3) Each trainee will have 15 minutes to complete the PC. 4) Denied assistance and references. 5) Must review Chapter 3, Enabling and Performance Checks and their definitions located in NPTORD 4500-54 prior to assessment of trainee.		
Assessment Limitations	Nil.	
Impact of Failure	Trainee will be given counselling with practice, a supplemental PC, and following additional failure TRB process will be initiated IAW NPTORD 4500-54	

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PROTECTED B (when complete)

PC 001 ENABLE SYSTEM FOR WEAPON USE			
Item	**SAT**	**UNSAT**	**Remarks**
MOUNT .50 CAL HMG M2HB QCB			
Manual override set to safe	☐	☐	**CRITICAL**
Place weapon on mount	☐	☐	
Place retaining pins in correct location	☐	☐	
Secure retaining pins	☐	☐	
CONVERT THE NRWS MOUNT FROM MANUAL TO REMOTE			
Manual override set to safe	☐	☐	**CRITICAL**
Disengage mount servos	☐	☐	
Disengage locking pins	☐	☐	
Disengage BLR	☐	☐	
Engage weapon safety	☐	☐	
CONVERT THE NRWS MOUNT FROM REMOTE TO MANUAL			
Disengage weapon safety	☐	☐	
Disengage BLR	☐	☐	
Engage mount servos	☐	☐	
Disengage locking pins	☐	☐	
Set MO to live	☐	☐	**CRITICAL**

Score: /14

Trainee must achieve 11/14 SATs INCLUDING ALL CRITICAL FACTORS to pass this assessment.

The trainee must achieve x/y SATs to pass this assessment.

Positive Feedback	**Improvement Feedback**

Overall Assessment: (Circle One)	**Met Standard**	**Did Not Meet Standard**

Assessor
I have discussed the results of this assessment with the trainee.

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PC 001 ENABLE SYSTEM FOR WEAPON USE		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

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PO 002 – PREPARE THE NRWS REMOTE OPERATING CONSOLE FOR USE

1. **Performance**: Prepare the NRWS Remote Operating Console (ROC) for use.

2. Conditions:

   a. Given:

      (1) Cue: Any necessary and required reason to prepare the NRWS ROC for use.
      (2) NRWS mount;
      (3) NRWS ROC
      (4) .50 Cal HMG M2HB QCB; and
      (5) C6A1 GPMG.

   b. Denied: Supervision; and

   c. Environment:

      (1) All weather conditions;
      (2) At sea or alongside;
      (3) Domestic or deployed ops; and
      (4) Day or night.

3. **Standard:** The ROCO shall, IAW A2, A3, A4, A5, A6, A7, A10, A11, A12, prepare the NRWS Remote Operating Console for use by:

   a. Performing system set-up by:

      (1) Identifying major components of the ROC;
      (2) Turning on system;
      (3) Performing Built In Test (BIT);
      (4) Conducting communications check;
      (5) Confirming status of FLIR;
      (6) Confirming Scan feature functionality;
      (7) Verifying Cross and Slave functions; and
      (8) Verifying weapon and ammunition settings.

   b. Making system mechanically and electrically safe by:

      (1) Making system mechanically safe by:
         (a) Removing the FEK; and
         (b) Engaging the Weapon Safety.
      (2) Making the system electrically safe by:

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(a) Observing and reacting to the Fire Inhibit Indicators by:
i. Noting the VETO indicator light;
ii. FEK light;
iii. MO light; and
iv. Gun Enable Switches.

c. Employing Modes of Operation by:

(1) Employing NRWS in Surveillance mode;
(2) Employing NRWS in Engagement mode;
(3) Employing NRWS in Offset mode;
(4) Employing NRWS in Home mode; and
(5) Employing NRWS in Service mode.

d. Converting NRWS ROC between remote to manual by:

(1) Setting MCP to Main;
(2) Selecting Home mode of operation; and
(3) Making NRWS mechanically safe by removing FEK.

4. **Training Limitations:** Nil.

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EO 002.01 – PERFORM SYSTEM SET UP
LG 002.01.01 – IDENTIFY MAJOR COMPONENTS OF ROC

LG 002.01.01 – Identify Major Components of ROC		
Location Conditions	**Instructional Sequence**	
Dedicated NRWS Classroom	After PC 001	
Time	**Classification**	**Instructor**
50 minutes	UNCLASS	RQ PO2 BOSN RQ S1 BOSN
Learning Objectives	**Instructional Method(s)**	
1) Identify major components of the ROC	1) EDI	
Scenario	A new junior member has been posted in to your Deck Department. Your task is to provide them with an overview of the NRWS mount.	
Activity Steps	1) Introduction (5 minutes) Instructor will introduce the ROC and its major components. 2) Performance/Activity (35 minutes – EDI) Instructor shall gather trainees at NRWS ROC for an interactive discussion to identify and explain the following major components: 1. MCP; 2. DU; 3. Auto record function and how to replay recordings; 4. FEK Assembly; and 5. MCU. 3) Conclusion (10 minutes) Instructor will have trainees come up individually to point out and explain various major components of the ROC. Instructor to assign homework assignment 002.01	
Instructor Resources	**References**	
1) Fully equipped NRWS mount 2) .50 Cal HMG M2HB QCB 3) Homework Assignment 4) NRWS Submenu layout poster	1) A2, A3, A4, A5, A6, A7, A10, A11, A12	
Note(s) to Instructor		
1) Instructor will emphasize critical safety importance of the manual override switch.		

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LG 002.01.02 – PERFORM SYSTEM SET UP

LG 002.01.02 – Perform System Set Up		
Location Conditions	**Instructional Sequence**	
NRWS Classroom	After LG 002.01.01	
Time	**Classification**	**Instructor**
50 minutes	UNCLASS	RQ PO2 BOSN RQ S1 BOSN
Learning Objectives	**Instructional Method(s)**	
1) Perform system set up of the ROC.	1) EDI	
Scenario	HMCS OROMOCTO is proceeding to sea and you are tasked to preform pre sail checks on the ROC.	
Activity Steps	1) Introduction (5 minutes) Instructor will introduce the basic functions required during the flash up procedures. 2) Performance/Activity (35 minutes – EDI) Instructor will utilize PPT to cover while students follow along at their computer based trainer (CBT) stations; 1. Turning on system; 2. Performing BIT; 3. Confirming status of FLIR; 4. Confirming Scan feature functionality; 5. Confirming Cross and Slave functions; and 6. Verifying weapon and ammunition settings 3) Conclusion (10 minutes) Trainees will individually demonstrate knowledge gained by performing actions above until entire class has done at least 1 step.	
Instructor Resources	**References**	
1) Fully equipped NRWS mount 2) .50 Cal HMG M2HB QCB 3) DWAN computer with projector 4) Power Point Presentation 5) NRWS Submenu layout poster 6) 1 x CBT per trainee	1) A2, A3, A4, A5, A6, A7, A10, A11, A12	
Note(s) to Instructor		
1) Intent is to make system set up walkthrough power point where all students are following along from their own stations as the instructor guides them. 2) CBT's are unable to conduct CROSS and SLAVE functions as they are stand alone units.		

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EO 002.02 – MAKE SYSTEM MECHANICALLY AND ELECTRICALLY SAFE
LG 002.02.01 – MAKE SYSTEM MECHANICALLY AND ELECTRICALLY SAFE

LG 002.02.01 – Make System Mechanically and Electrically Safe		
Location Conditions	**Instructional Sequence**	
Dedicated NRWS Classroom	After LG 002.02.01	
Time	**Classification**	**Instructor**
50 minutes	UNCLASS	RQ PO2 BOSN RQ S1 BOSN
Learning Objectives	**Instructional Method(s)**	
1) Make system mechanically safe 2) Make system electrically safe	1) EDI	
Scenario	You have been told to make the system mechanically and electrically safe as part of a demonstration of the system usage.	
Activity Steps	1) Introduction (10 minutes) Instructor will explain the process of making the system mechanically safe. 2) Performance/Activity (10 minutes – EDI) Instructor will demonstrate the steps required to make the system mechanically safe to include: 1. FEK removed; and 2. Weapon safety engaged. 3) Performance/Activity (5 minutes – Interactive Discussion) Instructor will demonstrate how to make the system electrically safe by observing and reacting to the VETO panels on board ship, MO switch, and fire inhibit indicators on DU. 4) Performance/Activity (5 minutes – EDI) Trainees will practice making the system electrically safe. 5) Performance/Activity (10 minutes – EDI) Trainees will make the system electrically safe by: 1. Weapon station gun enable switches in down position with cover closed; 2. MO switch to SAFE; and 3. Indicate on DU where the trainee would find the VETO status indication. 6) Conclusion (10 minutes) Instructor to emphasise the importance of making the system mechanically and electrically safe.	

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LG 002.02.01 – Make System Mechanically and Electrically Safe	
Instructor Resources	**References**
1) Fully equipped NRWS mount 2) .50 Cal HMG M2HB QCB 3) 1 x CBT per trainee 4) NRWS Submenu layout poster	1) A2, A3, A4, A5, A6, A7, A10, A11, A12
Note(s) to Instructor	
1) Instructor will emphasise critical safety importance of the manual override switch.	

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EO 002.03 – EMPLOY MODES OF OPERATION
LG 002.03.01 – EMPLOY MODES OF OPERATION

LG 002.03.01 – Employ Modes of Operation		
Location Conditions		**Instructional Sequence**
Dedicated NRWS Classroom		After LG 002.03.01
Time	**Classification**	**Instructor**
50 minutes	UNCLASS	RQ PO2 BOSN RQ S1 BOSN
Learning Objectives		**Instructional Method(s)**
1) Employ NRWS in Surveillance mode; 2) Employ NRWS in Engagement mode; 3) Employ NRWS in Offset mode; 4) Employ NRWS in Home mode; and 5) Employ NRWS in Service mode.		1) EDI
Scenario	A new junior member has been posted into your Deck Department. Your task is to provide them with an overview of the NRWS mount.	
Activity Steps	1) Introduction (5 minutes) Instructor shall give a brief description of the various modes of operation. 2) Performance/Activity (15 minutes – EDI) Instructor will EDI trainees around the mount to demonstrate the following: 1. Surveillance mode; 2. Engagement mode; 3. Offset mode; 4. Home mode; and 5. Service mode. 3) Performance/Activity (20 minutes) Trainees will practice selecting and using the various modes of operation while briefing the differences experienced. 4) Conclusion (10 minutes) Instructor to engage trainees by asking open ended questions (where's, why's, and how's) on the real life uses of these modes.	
Instructor Resources		**References**
1) Fully equipped NRWS mount 2) .50 Cal HMG M2HB QCB 3) 1 x CBT per trainee 4) NRWS Submenu layout poster		1) A2, A3, A4, A5, A6, A7, A10, A11, A12
Note(s) to Instructor		

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LG 002.03.01 – Employ Modes of Operation
1) Instructors shall ensure to relate the modes of operation to what the trainees will see on the DU so that the trainees can back brief during the practice phase.

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EO 002.04 – CONVERT NRWS ROC BETWEEN REMOTE TO MANUAL
LG 002.04.01 – CONVERT NRWS ROC BETWEEN REMOTE TO MAUAL

LG 002.04.01 – Convert NRWS ROC between Remote to Manual		
Location Conditions	**Instructional Sequence**	
Dedicated NRWS Classroom	After LG 002.04.01	
Time	**Classification**	**Instructor**
50 minutes	UNCLASS	RQ PO2 BOSN RQ S1 BOSN
Learning Objectives	**Instructional Method(s)**	
1) Convert ROC between remote and manual	1) EDI	
Scenario	Your ROC console has encountered a fault. You've been tasked to transfer the ROC to manual while troubleshooting takes place.	
Activity Steps	1) Introduction (5 minutes) Instructor will explain how to transfer the ROC between manual and remote 2) Performance/Activity (10 minutes – EDI) Instructor will demonstrate how to transfer between remote and manual by: 1. Setting MCP to Main; 2. Selecting Home mode of operation; and 3. Making NRWS mechanically safe by removing FEK. 3) Performance/Activity (30 minutes – EDI) Trainees will practice transferring the ROC between manual and remote. 4) Conclusion (5 minutes) Review LG and answer any relevant questions.	
Instructor Resources	**References**	
1) .50 Cal HMG M2HB QCB 2) Fully equipped NRWS mount 3) 1 x CBT per trainee 4) NRWS Submenu layout poster	1) A2, A3, A4, A5, A6, A7, A10, A11, A12	
Note(s) to Instructor		
1) Nil.		

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EC 002 – PREPARE THE NRWS REMOTE OPERATING CONSOLE FOR USE

EC 002 – Prepare the NRWS Remote Operating Console for Use.		
Scope	**Location Conditions**	**Instructional Sequence**
EO 002.01-EO 002.04	Dedicated NRWS Classroom	After LG 002.05.01
Time	**Classification**	**Instructor**
100 minutes	UNCLASS	RQ PO2 BOSN RQ S1 BOSN
Scenario	Trainees are closing up on the system to proceed to sea; must prove the weapon system correct.	
Assessment Steps	**Input** – Fully equipped NRWS trainer. **Performance** – IAW EC checklist. (12 minutes) **Output** – Trainee must attain 14/18 SATs INCLUDING ALL CRITICAL FACTORS to pass this assessment **Feedback** – Remaining time upon completion to be spent in group feedback IAW EC checklist.	
Instructor Resources		
1) Fully functioning NRWS trainer 2) .50 Cal HMG M2HB QCB		
Note(s) to Instructor		
1) Trainee's must mention how the VETO panel works and how it is shown on the DU.		
Assessment Limitations	Nil.	
Impact of Failure	Retrain/Reassess.	

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PROTECTED B (when complete)

PC 002 OPERATE THE NRWS REMOTE OPERATING CONSOLE			
Item	**SAT**	**UNSAT**	**Remarks**
PERFORM SYSTEM SET UP			
Turn on system	☐	☐	
Perform BIT	☐	☐	
Confirm status of FLIR	☐	☐	
Verify Cross and Slave functions	☐	☐	
Verify weapon and ammunition settings	☐	☐	
EMPLOY MODES OF OPERATION			
Employ NRWS in Surveillance mode	☐	☐	
Employ NRWS in Engagement mode	☐	☐	
Employ NRWS in Offset mode	☐	☐	
Employ NRWS in Home mode	☐	☐	
Employ NRWS in Service mode	☐	☐	
CONVERT NRWS ROC FROM REMOTE TO MANUAL			
Set MCP to Main	☐	☐	
Select Home mode of operation	☐	☐	
Make NRWS Mechanically Safe	☐	☐	**CRITICAL**
MAKE THE SYSTEM ELCTRICALLY SAFE			
Gun Enable Switches are in the down position and covers closed	☐	☐	**CRITICAL**
State Selector set to SAFE	☐	☐	**CRITICAL**
Gun Veto	☐	☐	**CRITICAL**
MAKE THE SYSTEM MECHANICALLY SAFE			
FEK removed	☐	☐	**CRITICAL**
Weapon Safety engaged	☐	☐	**CRITICAL**
Score: /18 **Trainee must achieve 14/18 SATs INCLUDING ALL CRITICAL FACTORS to pass this assessment.**			
Positive Feedback	**Improvement Feedback**		

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PC 002 OPERATE THE NRWS REMOTE OPERATING CONSOLE		
Overall Assessment: (Circle One)	**Met Standard**	**Did Not Meet Standard**
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

PO 002 – PREPARE THE NRWS REMOTE OPERATING CONSOLE FOR USE
PC 002 – PREPARE THE NRWS REMOTE OPERATING CONSOLE FOR USE

PC 002 – Prepare the NRWS Remote Operating Console for Use		
Scope	**Location Conditions**	**Instructional Sequence**
PO 002	Dedicated NRWS Classroom	After EC 002
Time	**Classification**	**Instructor**
150 minutes	UNCLASS	RQ PO2 BOSN RQ S1 BOSN
Scenario	A new junior member has been posted in to your Deck Department. Your task is to demonstrate to them how to operate the NRWS ROC.	
Assessment Steps	**Input** – Fully equipped NRWS trainer. **Performance** – IAW PC checklist. (18 minutes) **Output** – Trainee must attain 14/18 SATs INCLUDING ALL CRITICAL FACTORS to pass this assessment **Feedback** – Remaining time upon completion to be spent in group feedback IAW PC checklist.	
Instructor Resources		
1) Fully functioning NRWS trainer 2) .50 Cal HMG M2HB QCB		
Note(s) to Instructor		
1) Must review Chapter 3, Enabling and Performance Checks and their definitions located in NPTORD 4500-54 prior to assessment of trainee. 2) Trainee's must mention how the VETO panel works and how it is shown on the DU.		
Assessment Limitations	Nil.	
Impact of Failure	Trainee will be given counselling with practice, a supplemental PC, and following additional failure TRB process will be initiated IAW NPTORD 4500-54	

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PROTECTED B (when complete)

PC 002 OPERATE THE NRWS REMOTE OPERATING CONSOLE			
Item	**SAT**	**UNSAT**	**Remarks**
PERFORM SYSTEM SET UP			
Turn on system	☐	☐	
Perform BIT	☐	☐	
Confirm status of FLIR	☐	☐	
Verify Cross and Slave functions	☐	☐	
Verify weapon and ammunition settings	☐	☐	
EMPLOY MODES OF OPERATION			
Employ NRWS in Surveillance mode	☐	☐	
Employ NRWS in Engagement mode	☐	☐	
Employ NRWS in Offset mode	☐	☐	
Employ NRWS in Home mode	☐	☐	
Employ NRWS in Service mode	☐	☐	
CONVERT NRWS ROC FROM REMOTE TO MANUAL			
Set MCP to Main	☐	☐	
Select Home mode of operation	☐	☐	
Make NRWS Mechanically Safe	☐	☐	**CRITICAL**
MAKE THE SYSTEM ELCTRICALLY SAFE			
Gun Enable Switches are in the down position and covers closed	☐	☐	**CRITICAL**
State Selector set to SAFE	☐	☐	**CRITICAL**
Gun Veto	☐	☐	**CRITICAL**
MAKE THE SYSTEM MECHANICALLY SAFE			
FEK removed	☐	☐	**CRITICAL**
Weapon Safety engaged	☐	☐	**CRITICAL**
Score: /18 **Trainee must achieve 14/18 SATs INCLUDING ALL CRITICAL FACTORS to pass this assessment.**			
Positive Feedback	**Improvement Feedback**		

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PC 002 OPERATE THE NRWS REMOTE OPERATING CONSOLE		
Overall Assessment: (Circle One)	**Met Standard**	**Did Not Meet Standard**
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

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PO 003 – ENGAGE TARGETS WITH NRWS

1. **Performance**: Engage targets with NRWS.

2. **Conditions:**

   a. Given:

      (1) Cue: Dependant on the situation.
      (2) NRWS mount;
      (3) NRWS ROC;
      (4) .50 Cal HMG M2HB QCB;
      (5) C6A1 GPMG;
      (6) Ammunition;
      (7) Links; and
      (8) Linker/Delinker.

   b. Denied: Supervision; and

   c. Environment:

      (1) All weather conditions;
      (2) At sea or alongside;
      (3) Domestic or deployed ops; and
      (4) Day or night.

3. **Standard**: The NRWS Operator, IAW A2, A3, A4, A5, A6, A9, A10, A11, A12, shall engage targets with the Naval Remote Weapon System (CT0140, CT0150) by:

   a. Utilizing voice procedures.

   b. Conducting Weapon Handling Drills to include:

      (1) Close Up / Clear Away Procedures;
      (2) Loading NRWS;
      (3) Unloading NRWS;
      (4) Securing NRWS;
      (5) Standing Down NRWS;
      (6) Making Safe NRWS;
      (7) Performing Immediate Actions and Stoppages to include:
         (a) Reacting to initial misfire; and
         (b) Reacting to secondary misfire.
      (8) Deliberate Engagement Procedures;
      (9) Alarm Procedures;
      (10) Defensive Procedures;

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(11) Remote Optical Investigate;
(12) Remote Optical Scan; and
(13) Tracking Procedures.

4. **Training Limitations:** Nil.

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EO 003.01 – CONDUCT CLOSE UP/CLEAR AWAY PROCEDURES
LG 003.01.01 – CONDUCT CLOSE UP/CLEAR AWAY PROCEDURES

LG 003.01.01 – Conducting Close Up/Clear Away Procedures		
Location Conditions	**Instructional Sequence**	
Dedicated NRWS Classroom	After PC 002	
Time	**Classification**	**Instructor**
200 minutes	UNCLASS	RQ PO2 BOSN RQ S1 BOSN
Learning Objectives	**Instructional Method(s)**	
1) Utilize voice procedures 2) Conduct Weapon Handling Drills	1) EDI	
Scenario	Watch turnover is being conducted and you are required to conduct close up / clear away procedures.	
Activity Steps	1) Introduction (10 minutes – EDI) Instructor will demonstrate and explain the steps required to close up/clear away utilizing the voice procedures required. 2) Performance/Activity (20 minutes – EDI) Instructor will explain the new terms associated WRT to the weapon handling drills and procedures. 3) Performance/Activity (60 minutes – EDI) Instructor will utilize NCPM 1000 Annex C handout to demonstrate voice procedures for the NRWS. 4) Performance/Activity (100 min – EDI) Trainees will practice in teams of 2 conducting close up/clear away and voice procedures. 5) Conclusion (10 minutes) Instructor to reiterate the importance of voice procedures as trainees will now be expected to operate with these new terms and procedures.	
Instructor Resources	**References**	
1) Fully equipped NRWS mount 2) .50 Cal HMG M2HB QCB 3) 1 x CBT per trainee 4) NRWS Submenu layout poster 5) 1 x Handout per trainee	1) A2, A3, A4, A5, A6, A9, A10, A11, A12	
Note(s) to Instructor		
1) Instructor to maintain use of currently accepted drills and procedures as they are updated.		

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LG 003.01.01 – Conducting Close Up/Clear Away Procedures
2) NCPM 1000 Annex C in its entirety is to be printed and utilized as a handout for trainees.

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EO 003.02 – CONDUCT WEAPONS HANDLING DRILLS
LG 003.02.01 – CONDUCT WEAPONS HANDLING DRILLS

LG 003.02.01 – Conducting Weapons Handling Drills		
Location Conditions	**Instructional Sequence**	
Dedicated NRWS Classroom	After LG 003.01.01	
Time	**Classification**	**Instructor**
600 minutes	UNCLASS	RQ PO2 BOSN RQ S1 BOSN
Learning Objectives	**Instructional Method(s)**	
1) Conduct Weapon Handling Drills	1) EDI	
Scenario	HMCS LONDON is conducting a deliberate shoot and as ROC Operator you are tasked to practice drills.	
Activity Steps	1) Introduction (5 minutes) Instructor will introduce the new weapon handling drills associated with the NRWS. 2) Performance/Activity (5 minutes – EDI) Instructor will review NCPM 1000 Annex C with students to include: 1. Command Terms and Definitions; 2. Positive Weapon Control; 3. Weapon State Policy; and 4. States of Preparedness. 3) Performance/Activity (60 minutes – EDI) Instructor will demonstrate drills to include: 1. Load; 2. Unload; 3. Make Safe; 4. Immediate Action; 5. Stoppage After IA; 6. Deliberate Engagement Procedures; 7. Alarm Procedures; 8. Defensive Procedures; 9. Remote Optical Investigate; 10. Remote Optical Scan; and 11. Tracking Procedures. 4) Performance/Activity (470 minutes – EDI) Trainees will practice NRWS drills in pairs utilizing CBT's, embedded training scenarios, and weapon system simulator (WSS).		

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LG 003.02.01 – Conducting Weapons Handling Drills	
5) Conclusion (10 minutes) Instructor will reinforce the need to continue practicing once trainees are on ship to maintain proficiency.	
Instructor Resources	**References**
1) Fully equipped NRWS mount 2) .50 Cal HMG M2HB QCB 3) 1 x CBT per trainee 4) Embedded scenarios 5) Weapon Handling Drills Handout 6) Linker/Delinker 7) WSS 8) Practice Ammunition 9) NRWS Submenu layout poster 10) Weapon State Policy orders printed and laminated for each station	1) A2, A3, A4, A5, A6, A9, A10, A11, A12
Note(s) to Instructor	
1) Instructor to explain what critical factors are and the implication of failing critical items. 2) NCPM 1000 Annex C in its entirety is to be printed and utilized as a handout for trainees.	

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EC 003 – ENGAGE TARGETS WITH NRWS

EC 003 – Engage Targets with NRWS		
Scope	**Location Conditions**	**Instructional Sequence**
EO 003.01-EO003.02	Dedicated NRWS Classroom	After LG 003.02.01
Time	**Classification**	**Instructor**
250 minutes	UNCLASS	RQ PO2 BOSN RQ S1 BOSN
Scenario	Your ship will be conducting CRR 2-54 for NRWS Surface Engagement and you have been designated as the ROCO for the firing.	
Assessment Steps	**Input** – Fully equipped NRWS trainer. **Performance** – IAW EC checklist. (14 minutes) **Output** – Trainee must achieve 19/27 SATs INCLUDING ALL CRITICAL FACTORS. **Feedback** – Immediate feedback upon completion (2 minutes)	
Instructor Resources		
1) Fully functional NRWS trainer 2) .50 Cal HMG M2HB QCB 3) Practice Rounds 4) Links 5) Linker/Delinker		
Note(s) to Instructor		
1) Trainees are to complete in teams, then switch roles until all trainees have completed both tests. 2) Loaders will not be tested as part of this EC, however they will be expected to conduct themselves fully in their position IOT ensure the testing occurs in its entirety. 3) Test will consist of a Deliberate Engagement Procedure 4) Must review Chapter 3, Enabling and Performance Checks and their definitions located in NPTORD 4500-54 prior to assessment of trainee. 5) Instructor/Invigilator to act in the roles of WDO, CO, and DSO as necessary.		
Assessment Limitations	Nil.	
Impact of Failure	NPTORD 4500-54.	

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EC 003 Engage Targets with NRWS			
Item	**SAT**	**UNSAT**	**Remarks**
CONDUCTS CLOSE UP/CLEAR AWAY			
Establish Communications	☐	☐	
Set State Selector To Safe	☐	☐	CRITICAL
Set Gun Enable Switches are in down position	☐	☐	CRITICAL
Ensures FEK removed	☐	☐	CRITICAL
Reports	☐	☐	
LOAD			
Enters the ammo amount in MFD	☐	☐	
Inserts and enables FEK	☐	☐	
Reports	☐	☐	
ENGAGES TARGET			
Slews to ordered bearing	☐	☐	
Selects LRF	☐	☐	
Selects Engage Mode	☐	☐	
Set Gun Enable Switches in up position	☐	☐	
Selects Charge	☐	☐	
Selects Arm	☐	☐	
Locks on with EOSS	☐	☐	
Selects Burst Lo	☐	☐	
Reports	☐	☐	
Depresses Firing Button	☐	☐	
Observes fall of shot	☐	☐	
Makes corrections	☐	☐	
Engages until target destroyed	☐	☐	
Reports	☐	☐	
CEASE FIRE / UNLOAD			
Slews To Ordered Bearing / Home Position	☐	☐	
State Selector To Safe	☐	☐	CRITICAL
Gun Enable Switches To Down Position	☐	☐	CRITICAL
FEK To Disable	☐	☐	CRITICAL
Reports	☐	☐	
Score: /27			

EC 003 Engage Targets with NRWS		
Trainee must achieve 19/27 SATs INCLUDING ALL CRITICAL FACTORS to pass this assessment.		
Positive Feedback	**Improvement Feedback**	
Overall Assessment (Circle One)	**Met Standard**	**Did Not Meet Standard**
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

PO 003 – ENGAGE TARGETS WITH NRWS
PC 003 – ENGAGE TARGETS WITH NRWS

PC 003 – Engage Targets with NRWS		
Scope	**Location Conditions**	**Instructional Sequence**
PO 003	Dedicated NRWS Classroom	After EC 003
Time	**Classification**	**Instructor**
250 minutes	UNCLASS	RQ PO2 BOSN RQ S1 BOSN
Scenario	Your ship will be conducting CRR 2-54 for NRWS Surface Engagement and you are designated ROCO for the firing.	
Assessment Steps	**Input** – Fully equipped NRWS trainer. **Performance** – IAW PC checklist. (14 minutes) **Output** – Trainee must achieve 19/27 SATs INCLUDING ALL CRITICAL FACTORS. **Feedback** –Immediate feedback upon completion. (2 minutes)	
Instructor Resources		
1) Fully functional NRWS 2) .50 Cal HMG M2HB QCB 3) Practice Rounds 4) Links 5) Linker/Delinker		
Note(s) to Instructor		
1) Trainees are to complete in teams, then switch roles until all trainees have completed both tests. 2) Loaders will not be tested as part of this EC, however they will be expected to conduct themselves fully in their position IOT ensure the testing occurs in its entirety 3) Test will consist of an Alarm Procedure 4) Must review Chapter 3, Enabling and Performance Checks and their definitions located in NPTORD 4500-54 prior to assessment of trainee. 5) Instructor/Invigilator to act in the roles of WDO, CO, and DSO as necessary.		
Assessment Limitations	Nil.	
Impact of Failure	Trainee will be given counselling with practice, a supplemental PC, and following additional failure TRB process will be initiated IAW NPTORD 4500-54	

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PROTECTED B (when complete)

PC 003 OPERATE NRWS			
Item	**SAT**	**UNSAT**	**Remarks**
CONDUCT CLOSE UP/CLEAR AWAY			
Establish Comms	☐	☐	
State Selector To Safe	☐	☐	CRITICAL
Gun Enable Switches Down Position	☐	☐	CRITICAL
Ensures FEK removed	☐	☐	CRITICAL
Reports	☐	☐	
LOAD			
Enters The Ammo Amount In MFD	☐	☐	
Inserts And Enables FEK	☐	☐	
Reports	☐	☐	
ENGAGES TARGET			
Slews To Ordered Bearing	☐	☐	
Selects LRF	☐	☐	
Selects Engage Mode	☐	☐	
Gun Enable Switches In Up Position	☐	☐	
Selects Charge	☐	☐	
Selects Arm	☐	☐	
Locks On With EOSS	☐	☐	
Selects Burst Lo	☐	☐	
Reports	☐	☐	
Depresses Firing Button	☐	☐	
Observes Fall Of Shot	☐	☐	
Makes Corrections	☐	☐	
Engages Until Destroyed	☐	☐	
Reports	☐	☐	
CEASE FIRE/UNLOAD			
Slews To Ordered Bearing / Home Mode	☐	☐	
State Selector To Safe	☐	☐	CRITICAL
Gun Enable Switches To Down Position	☐	☐	CRITICAL
FEK To Disable	☐	☐	CRITICAL
Reports	☐	☐	
Score: /27			

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PC 003 OPERATE NRWS		
Trainee must achieve 19/27 SATs INCLUDING ALL CRITICAL FACTORS to pass this assessment.		
Positive Feedback	**Improvement Feedback**	
Overall Assessment: (Circle One)	**Met Standard**	**Did Not Meet Standard**
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

PO 004 – CORRECT MINOR NRWS FAULTS

1. **Performance:** Correct minor NRWS faults.

2. **Conditions:**

   a. Given:

      (1) Cue: Issue encountered with system or directed to complete or assist in clearing a located issue.
      (2) NRWS mount;
      (3) NRWS ROC;
      (4) .50 Cal HMG M2HB QCB;
      (5) C6A1 GPMG;
      (6) Ammunition;
      (7) Links; and
      (8) Linker/Delinker.

   b. Denied: Supervision; and

   c. Environment:

      (1) All weather conditions;
      (2) At sea or alongside;
      (3) Domestic or deployed ops; and
      (4) Day or night.

3. **Standard:** The NRWS Operator, IAW A3, A4, A5, A6, A7, A8, A9, A10, A11, A12, shall correct minor NRWS faults by:

   a. Responding to mount malfunctions to include:

      (1) Gun charging malfunction; and
      (2) Automatic tracking failure.

   b. Responding to stabilization faults to include:

      (1) Unreasonable elevation and/or azimuth readings; and
      (2) Weapon station does not move in azimuth or elevation.

   c. Responding to ROC faults to include:

      (1) System not entering service mode;
      (2) No video display;
      (3) System not entering engage mode; and

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(4) No communication with MCU.

4. **Training Limitations:** Nil.

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EO 004.01 – RESPOND TO MOUNT MALFUNCTIONS
LG 004.01.01 – RESPOND TO MOUNT MALFUNCTIONS

LG 004.01.01 – Respond to Mount Malfunctions		
Location Conditions	**Instructional Sequence**	
Dedicated NRWS Classroom	After PC 003	
Time	**Classification**	**Instructor**
50 minutes	UNCLASS	RQ PO2 BOSN RQ S1 BOSN
Learning Objectives	**Instructional Method(s)**	
1) Respond to mount malfunctions	1) EDI	
Scenario	Your NRWS is experiencing issues. You need to investigate, diagnose and resolve minor faults.	
Activity Steps	1) Introduction (5 minutes – EDI) Instructor will explain how mount malfunctions can occur. 2) Performance/Activity (10 minutes – EDI) Instructor will demonstrate how to resolve faults at the mount to include: 1. Gun charging malfunction; and 2. Automatic tracking failure. 3) Performance/Activity (30 minutes – EDI) Trainees will resolve mount malfunctions in pairs. 4) Conclusion (5 minutes) Instructor shall emphasize the importance of clearing mount malfunctions during all engagement scenarios.	
Instructor Resources	**References**	
1) Fully equipped NRWS mount 2) .50 Cal HMG M2HB QCB 3) NRWS Submenu layout poster	1) A3, A4, A5, A6, A7, A8, A9, A10, A11, A12	
Note(s) to Instructor		
1) Instructor will emphasize critical safety importance of the manual override switch		

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EO 004.02 – RESPOND TO STABILIZATION FAULTS
LG 004.02.01 – RESPOND TO STABILIZATION FAULTS

LG 004.02.01 – Responding to Stabilization Faults		
Location Conditions	**Instructional Sequence**	
Dedicated NRWS Classroom	After LG 004.01.01	
Time	**Classification**	**Instructor**
50 minutes	UNCLASS	RQ PO2 BOSN RQ S1 BOSN
Learning Objectives	**Instructional Method(s)**	
1) Respond to stabilization faults	1) EDI	
Scenario	Your NRWS is experiencing issues. You need to investigate, diagnose and resolve minor faults.	
Activity Steps	1) Introduction (10 minutes – EDI) Instructor shall explain how stabilization faults can occur. 2) Performance/Activity (10 minutes – EDI) Instructor shall demonstrate how to resolve: 1. Unreasonable elevation and/or azimuth readings; and 2. Weapon station does not move in azimuth or elevation. 3) Performance/Activity (20 minutes – EDI) Trainees will practice resolving stabilization faults in pairs. 4) Conclusion (10 minutes) Instructor shall emphasize the importance of clearing stabilization faults during all engagement scenarios.	
Instructor Resources	**References**	
1) Fully equipped NRWS mount 2) .50 Cal HMG M2HB QCB 3) NRWS Submenu layout poster	1) A3, A4, A5, A6, A7, A8, A9, A10, A11, A12	
Note(s) to Instructor		
1) Instructor will emphasize critical safety importance of the manual override switch.		

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EO 004.03 – RESPOND TO ROC FAULTS
LG 004.03.01 – RESPOND TO ROC FAULTS

LG 004.03.01 – Responding to ROC Faults		
Location Conditions		**Instructional Sequence**
Dedicated NRWS Classroom		After LG 004.02.01
Time	**Classification**	**Instructor**
50 minutes	UNCLASS	RQ PO2 BOSN RQ S1 BOSN
Learning Objectives		**Instructional Method(s)**
1) Respond to ROC Faults		1) EDI
Scenario	Your NRWS is experiencing issues. You need to investigate, diagnose and resolve minor faults.	
Activity Steps	1) Introduction (5 minutes) Instructor will describe how the ROC can encounter faults. 2) Performance/Activity (10 minutes – EDI) Instructor will explain and demonstrate how to identify and resolve the following: 1. System not entering service mode; 2. No video display; 3. System not entering engage mode; and 4. No communication with MCU. 3) Performance/Activity (30 minutes – EDI) Trainees will practice resolving ROC faults in pairs. 4) Conclusion (5 minutes) Instructor shall emphasize the importance of clearing ROC faults during all engagements.	
Instructor Resources		**References**
1) 1 x CBT per trainee 2) Fully equipped NRWS mount 3) .50 Cal HMG M2HB QCB 4) NRWS Submenu layout poster		1) A3, A4, A5, A6, A7, A8, A9, A10, A11, A12
Note(s) to Instructor		
1) Instructor will emphasize critical safety importance of the manual override switch.		

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EC 004 – CORRECT MINOR NRWS FAULTS

EC 004 – Correct Minor NRWS Faults		
Scope	**Location Conditions**	**Instructional Sequence**
EO 004.01-EO 004.03	Dedicated NRWS Classroom	After LG 004.03.01
Time	**Classification**	**Instructor**
50 minutes	UNCLASS	1 x RQ PO2 BOSN 1 x RQ S1 BOSN
Scenario	During preparations for a deliberate firing, your ROC experiences several minor malfunctions. You must troubleshoot the issues in order to carry on with the firing.	
Assessment Steps	**Input** – Fully equipped NRWS trainer. **Performance** – IAW EC checklist. (6 minutes) **Output** – Trainee must attain 3/4 SAT to pass this assessment **Feedback** – If time permits, group feedback upon completion, IAW EC checklist.	
Instructor Resources		
1) Fully functional NRWS 2) .50 Cal HMG M2HB QCB 3) Practice rounds 4) Links 5) Linker/Delinker		
Note(s) to Instructor		
1) Nil.		
Assessment Limitations	Nil.	
Impact of Failure	Retrain/Reassess.	

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PROTECTED B (when complete)

EC 004 Correct Minor NRWS FAULTS			
Item	**SAT**	**UNSAT**	**Remarks**
RESPOND TO MOUNT MALFUNCTIONS			
Gun Charging Malfunction	☐	☐	
RESPOND TO STABILIZATION FAULTS			
Weapon Station Does Not Move In Azimuth Or Elevation	☐	☐	
RESPOND TO ROC FAULTS			
System Not Entering Engage Mode	☐	☐	
Automatic Tracking Failure	☐	☐	

Score: /4

The trainee must achieve 3/4 SATs to pass this assessment.

Positive Feedback	Improvement Feedback

Overall Assessment (Circle One)	Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

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PO 004 – CORRECT MINOR NRWS FAULTS
PC 004 – CORRECT MINOR NRWS FAULTS

PC 004 – Correct Minor NRWS Faults		
Scope	**Location Conditions**	**Instructional Sequence**
PO 004	Dedicated NRWS Classroom	After EC 004
Time	**Classification**	**Instructor**
50 minutes	UNCLASS	RQ PO2 BOSN RQ S1 BOSN
Scenario	During preparation for a deliberate firing, your ROC experiences several minor malfunctions. You must troubleshoot the issues in order to carry one with the firing.	
Assessment Steps	**Input** – Fully equipped NRWS trainer. **Performance** – IAW PC checklist. (6 minutes) **Output** – Trainee must attain 2/3 SAT to pass this assessment **Feedback** – If time permits, group feedback upon completion IAW PC checklist.	
Instructor Resources		
1) Fully functional NRWS 2) .50 Cal HMG M2HB QCB 3) Practice rounds 4) Links 5) Linker/Delinker		
Note(s) to Instructor		
1) Must review Chapter 3, Enabling and Performance Checks and their definitions located in NPTORD 4500-54 prior to assessment start.		
Assessment Limitations	Nil.	
Impact of Failure	Trainee will be given counselling with practice, a supplemental PC, and following additional failure TRB process will be initiated IAW NPTORD 4500-54	

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PROTECTED B (when complete)

PC 004 CORRECT MINOR NRWS FAULTS			
Item	**SAT**	**UNSAT**	**Remarks**
RESPOND TO MOUNT MALFUNCTIONS			
Gun Charging Malfunction	☐	☐	
RESPOND TO STABILIZATION FAULTS			
Weapon Station Does Not Move In Azimuth Or Elevation	☐	☐	
RESPOND TO ROC FAULTS			
System Not Entering Engage Mode	☐	☐	

Score: /3

The trainee must achieve 2/3 SATs to pass this assessment.

Positive Feedback	Improvement Feedback

Overall Assessment: (Circle One)	**Met Standard**	**Did Not Meet Standard**
Assessor I have discussed the results of this assessment with the trainee.		
Name:	Date:	Signature:
Trainee I have been debriefed on the result of this assessment.		
Name:	Date:	Signature:

PROTECTED B (when complete)

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CHAPTER 3: ASSESSMENT OF TRAINEE

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GENERAL

1. All PCs are based on POs in Chapter 2 of this QSP. Trainees must achieve the standard to be awarded the qualification. PCs will be administered in the form of practical or written assessments and the trainee must pass all POs to successfully complete this course.

ENABLING AND PERFORMANCE CHECKS

2. All ECs are formative assessments that help the trainee and the instructors to recognize progress or gaps in learning so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. ECs also reinforce learning and increase the retention of material for longer periods in preparation for the PC. Non-critical EC results do not factor into a trainee's successful completion of the PO.

3. The process for critical ECs and PCs is outlined in NPTORD 4500-54. For administrative purposes, a Critical EC has the same importance as a PC.

4. PCs are summative assessments used to determine if the trainee has met the standard under the conditions (where possible) of the PO. Supplemental PCs are administered upon failure of a PC and must be in the same format and have the same standard as the original PC.

RESPONSIBILITY FOR TESTING

5. Assessment of trainees is to be completed IAW the standards delineated in this QSP, NPTORD 4500-54, and CFITES Vol 7 – Evaluation.

PROGRESS MONITORING

6. Monitoring learning is the responsibility of all instructional staff and supervisors. Monitoring, including unsatisfactory progress, shall adhere to CFITES Vol 6 – Conduct of Instructional Programmes.

7. Continuous monitoring of trainee progress is required to provide the following:

8. Early warning of trainee difficulties;

9. Feedback on the effectiveness of training;

10. Information for the Training Review Board (TRB) (if required); and

11. Information to assist in the completion of the CF 377 Course Report.

12. Trainees experiencing difficulty in any area of performance are to be counselled with regard to the nature of their shortcomings and suggested corrective action IAW TE SOs/SOPs and DAOD 5516-5.

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13. PC failures and supplemental PC failures are to be handled IAW TE SOs/SOPs and NPTORD 4500-54. Supplemental ECs and PCs shall not be administered within 24 hours of the trainee being informed of their EC or PC failure. Trainees will have 24 hours at minimum to prepare for a retest.

TRAINING REVIEW BOARD

14. The aim of the TRB process is to review a trainee's suitability for continued training and education, while ensuring that the trainee is afforded a fair and just assessment of all relevant information before a decision is made. TRBs shall be convened and conducted IAW NPTORD 4500-54.

COURSE REPORTS

15. Trainees shall be graded pass or fail IAW NPTORD 4500-51.

16. Course reports (CF 377) shall be prepared and distributed IAW NPTORD 4500-51. If applicable, training limitations and/or incomplete training must be included in the course report.

17. Narratives shall indicate trainee's strengths and weaknesses based on PC results to provide information to the Career Manager.

18. A record of trainee counselling shall be maintained for 2 years by the TE IAW Information Management Standard (IMS) 6001-2-1 and Defence Subject Classification and Disposition System (DSCDS) Primary Number 4500.

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ANNEX A – REFERENCES

This table lists all references used within this QSP.

#	Short Title	Long Title
A – CAF or other Official Canadian Government Publications		
A1	NCPM 704	NCPM 704 50 Calibre Heavy Machine Gun Drills and Procedures Manual
A2	NCPM 1000 V7 Annex C	NCPM 1000 Version 7.0 Annex C Naval Remote Weapons Station (NRWS)
A3	System User Manual	C-70-411-000/MB-001 System User Manual Naval Remote Weapons Station (NRWS)
A4	Operations Instructions	C-70-411-000/MB-001 Operation Instructions Naval Remote Weapon Station (NRWS) System
A5	Installation/Assembly Instructions	C-70-411-000/ME-005 Installation/Assembly Instructions Naval Remote Weapon Station (NRWS) System
A6	Installation and User Manual	C-70-411-000/ME-003 Installation and User Manual Naval Remote Weapon Station (NRWS) Computer Based Trainers (CBT)
A7	Description and Preventative Maintenance Instructions	C-70-411-000/MF-001 Description and Preventative Maintenance Instructions Naval Remote Weapon Station (NRWS) System
A8	Description and Preventative Maintenance Instructions	C-70-411-000/MF-002 Description and Preventative Maintenance Instructions Naval Remote Weapon Station (NRWS) Electro-Optical Payload (MINIPOP)
A9	Maintenance Instructions	C-70-411-000/MM-001 Maintenance Instructions Naval Remote Weapon Station (NRWS) System
A10	Illustrated Parts Breakdown	C-70-411-000/MY-001 Illustrated Parts Breakdown Naval Remote Weapon Station (NRWS) System
A11	System Specification	C-70-411-000/SF-001 System Specification Naval Remote Weapon Station (NRWS) System
A12	SSO’s AL 11	Royal Canadian Navy Ships Standing Orders Al 11
B – Official Allied Force and Allied Government Publications		
Nil		
C – Commercial Publications		
Nil		
D – Other Publications		
Nil		

ANNEX B – TASK LIST

This table provides an overview of MTL tasks WRT if/where they are trained within this QSP. No train substantiations are located in the Record of Decisions annex.

Task #	Task Statement	T/NT	PO
CT0140, CT0150	The NRWS Operator, IAW references, shall engage targets with the Naval Remote Weapon System (NRWS)	T	003
Added Tasks			
Nil	Enable System For Weapon Use	T	001
Nil	The ROCO shall prepare the NRWS Remote Operating Console for use	T	002
Nil	The NRWS Operator shall Correct Minor NRWS Faults	T	004

ANNEX C – RECORD OF DECISIONS

The following summary provides insight to the decisions and reasoning resulting in this QSP.

Dates: 27-29 Sept and 1 Oct 2021

Location: S120, Stadacona, CFB Halifax

Working Group Members:

Surname	Initials	Rank	MOSID	Unit
Allison	S.R.	LT(N)	00207-06	NTDC(A)
MacMullin	A.	Capt	00211-02	NTDC(A)
Leblanc	I.D.	SLt	0211-02	NTDC(A)
Handley	H.	Capt	00211-02	NPTG HQ
Kreger	A.	CPO2	00105	NPTG HQ
Skiffington	J.	CPO2	00105	NTDC(A)
Hann	L.	CPO2	00105	ST(A)
Conway	J.	CPO2	00105	NFR
Murray	M.	PO1	00115	ST(A)
Summerfield	C.	PO2	00105	NTDC(A)
Fitzmorris	D.	PO2	00105	NTDC(A)
Stewart	A.	PO2	00105	NTDC(A)
MacDonald	B.	MS	00366-05	NFSA
Perry	P.	S1	00105	NFSA
Myles	A.	S1	00105	NFSA

Working Group Objective:

19. The purpose of this working group was to develop the initial training QSP for the NRWS. This QSP would be used on both coasts, fleet wide to train a large portion of personnel in the use of the NRWS.

Specification Task – No Train Decisions:

Task #	Task Statement	Substantiation
Nil	All tasks are to be trained	As this is a new QSP, all tasks will be trained at this time.

Discussions / Decisions:

DAY	DISCUSSION/ DECISION (S)	COMMENTS

Day 1	WG discussed what other tasks needed to be train: i.e., MOB, FP, and NBP Surface (Support) Engage Targets. WG stated that a runaway gun in remote and manually would be same as in 50 Cal course. WG discussed when to include in RQ Course. WG's CPO2 Hann recommended that NRWS training be added to end of 50 Cal op course. WG noted that C6 and 50 Cal would be pre requisites for NRWS Trg. WG's Capt MacMullin gave brief on WG process and CFITES. WG PO1 Murray / CPO2 Hann suggested that two Performance Objectives are req'd: Load NRWS, Operate ROCO. WG agreed that supervisors require extra training but still need initial NRWS training to know the system better than subordinates. WG discussed making QSP Modular. WG discussed the following potential NRWS tasks and others: unload, miss fire crewing, any Nav Comm involvement, ST issue, Turn camera on, train / elevate, pair mounts ,camera function, must list all tasks, no matter how small, decided to turn on even though CSE will likely have already done it. WG discussed emergency procedures - blackout? WG agrees course should have ship tour for system. WG agrees care and control of FEK is CSO's IAW NAVORD. WG discussed what to do with ejected round, how to identify stoppages where to witness from, charge it, doesn't fire second time, send someone out to check.	

	WG CPO2 Conway expressed concern about round piling up on breach block and recommended safety blurbs be added in about breach impeded. S1 Myles asked if stated that loader witness round expelled and PO1 Murray recommend inspect after second consecutive misfire. WG Board Chair determined this discussion is not applicable to working group, will be addressed after through drills. WG discussed whether or not ROCO makes electrically safe or mechanically safe for clearing secondary IA. WG discussed Home Position must be declared safe bearing before slewing weapon to it. There was debate over Home Position vs safe bearing and confirming report of initial misfire to WDO IAW NCPM 1000. WG was to confirm Mech vs Elec safe IAW NAVORD. WG noticed that Mech and Elec safe not defined in NAVORD. Definitions from Interim Guidance will be used. WG agree that NCPM Drills be improve in NCPM 1000 including: Offset in system an issue; WDO to give W.S. bearing; who gets the training; when do we create supervisor portion WG agreed that NRWS is trained to CPF as core training. WG discussed the following: When do BOSNs learn policy? Do they need it? Understand function, capabilities; add in supervisor mod after; NRWS training geared toward S1 only.	

Day 2	WG determined that 6 new POs were from email. WG performed Task analysis. WG added task “identify react to initial misfire/secondary misfire”. WG change MFD name to ROC. WG agreed that change system overview will cover explain and introduce system overview and description; transfer manual to remote and back; put in operate NRWS mount. WG agreed 1 period guest lecture from NFS CSE to go over weapon change over. WG disagree on whether or not to train BIT testing for Operators, belongs to CSE Tech WG reached a quorum on scalar achieved. WG released PO1 Murray services and thanked him for contributions. WG explain system overview should be written test but TDO said it must be EC if written test.	
Day 3	WG began discussion on policy & procedures noting that policy is NT as it can change, will be given by instructor but not taught - CPO2 Conway, (West Coast TDO) WG acknowledge NCPM 1000 has drills - load, make safe, reload, IA's & Stoppages only. WG agreed on PO 003 title - Operate NRWS. WG agreed on PO 004 - Correcting NRWS Faults include: screen faults, train & elevate weapon changed to stabilization fails, no video display, gun charge malfunction, gun does not fire. WG agreed scalar and POs complete & agreed on @ 1030 WG released MS MacDonald (CSE SME), thanked him for his expertise.	

	WG started PC creation began @ 1100 and agreed on the following items: mounting C6 for testing because 50 Cal will be assets later; 70% for a pass to be confirmed via references. WG happy with pass mark of 7O% of assessed PC items with the exception of S1 Myles. WG discussed NRWS tasks into JTAR. WG discussed 50 Cal mounting not tested as it will be most used & mounted. WG agreed on PC 001 & PC 002 completed. WG to used CFCD 106 to be referred to for unclassified close up/clear away procedures. WG decided load & unload left as 1 step as drills are assessed in prerequisite training. WG decided that load & unload - MO to safe added to both as critical factor. WG divided PC 003 broken into 2 parts (A & B). WG decided to add drills found in NCPM 1000 used in PC 003.	
Day 4	WG agreed on PC 003 complete. WG agreed assessment steps to be filled in post W.G. and test total TBD post W.G. WG started building PC 004 and determined most common and complex NRWS faults to be utilized. WG decoded Checklist steps to be input post W.G. and test total TBD post W.G. WG agreed pm PC 004 completed. WG began work on QSP Chapter 1 WG TDO gave brief on Lesson Guidance, EO's, and LO's.	

	WG Began work on Chapt 2 – EO's to be completed post WG	
27 April 2022	Mbrs present were: PO 2 Summerfield, Bosn, NTDC (A) PO 2 Hayes, Bosn, NTDC (A) Slt Leblanc, TDO, NTDC (A) Decision was made to alter the titles of PO's 1, 2, and 3 IOT make their requirements more clearly defined. As well it was decided to leave no task numbers to PO's 1, 2, and 4 as they do not have a number within the JTAR and numbers provided (engage targets) did not accurately reflect performances of these POs. TDO on behalf of the board recommended to Occ Mgr to create more task numbers to reflect new core tasks associated with NRWS-specific functions but was not granted recommendation. Will proceed with CT0140 and CT0150 as the only numbered tasks in this QSP.	
18 May 2022	The following changes were made: all AIT was removed including the AIT manual as a reference. Identifying components were separate EO's; they were relocated as LG's into follow on EO's. Course Duration was recalculated as it was faulty. Cue's were added to all PO's. Environmental conditions were changed to read "All environmental conditions". Test titles were changed to better reflect proper overall requirements.	
27 May 2022	Formatting corrections were made. EC's and PC's for PO's 001, 002, 003, and 004 were improved to better suit process over performance.	

20.
Follow Up Action:

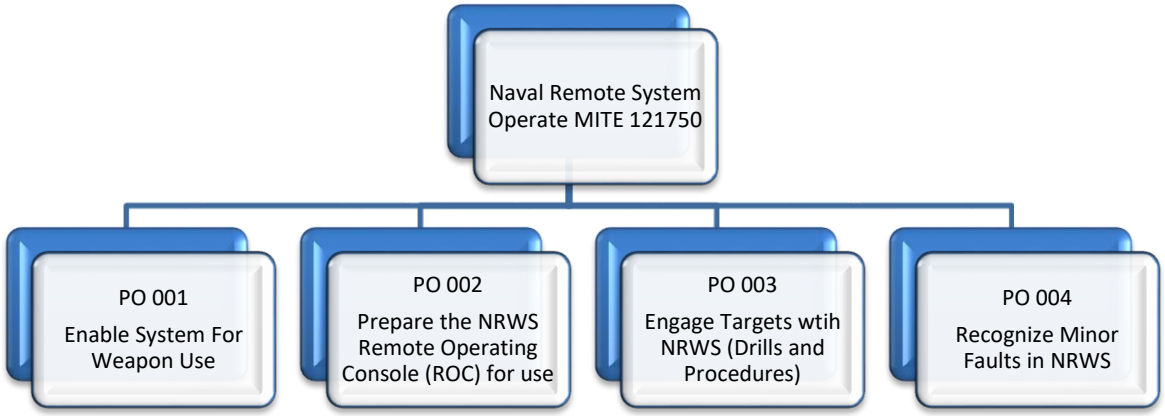
Item	Action	OPI	Expected Date of Completion
JTAR	Bosn JTAR does not include NRWS as a listed item. Tasks listed in ref A require task numbers assigned and must be added to JTAR.	TM	TBD
NCPM 1000 Annex C	Once NCPM 1000 Annex C is updated with NRWS drills and procedures, the NRWS QSP needs to be amended to reflect to align those changes with the RNWS QSP performances.	PO 2 Summerfield	TBD

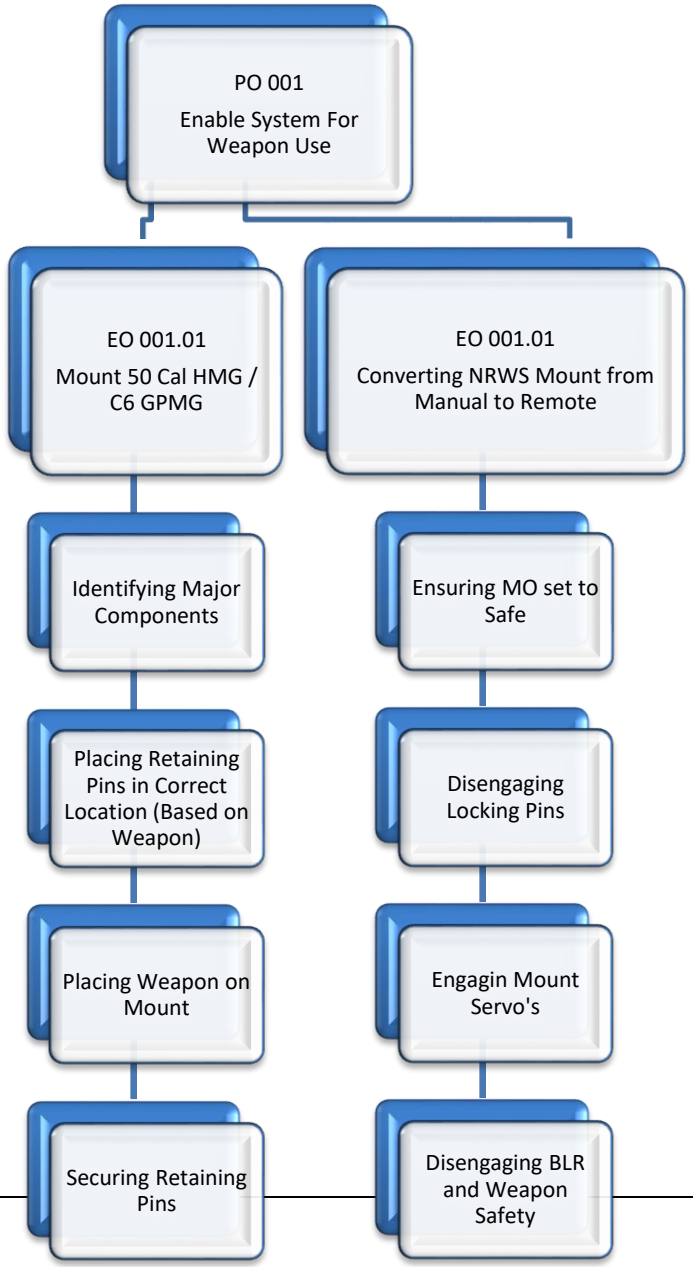
Course History:

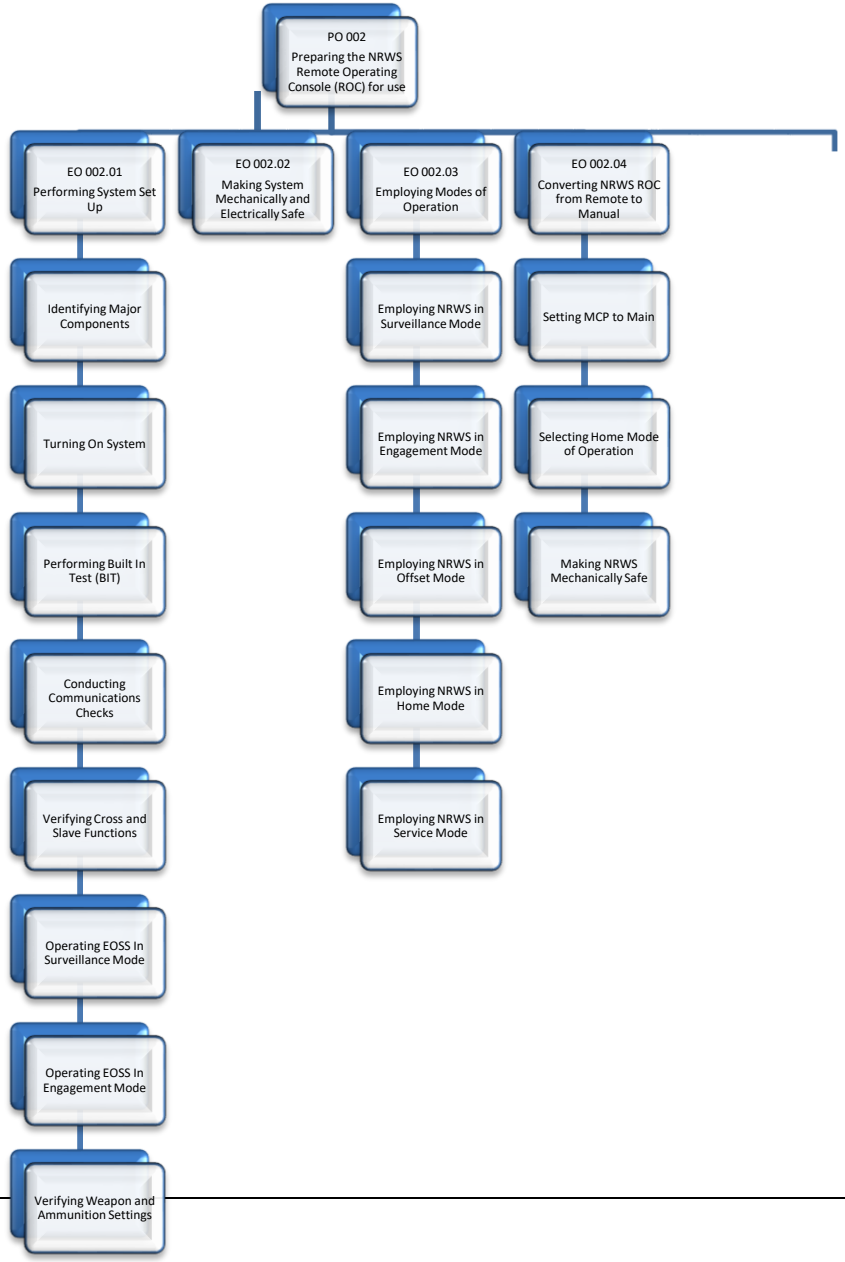
21. Nil.

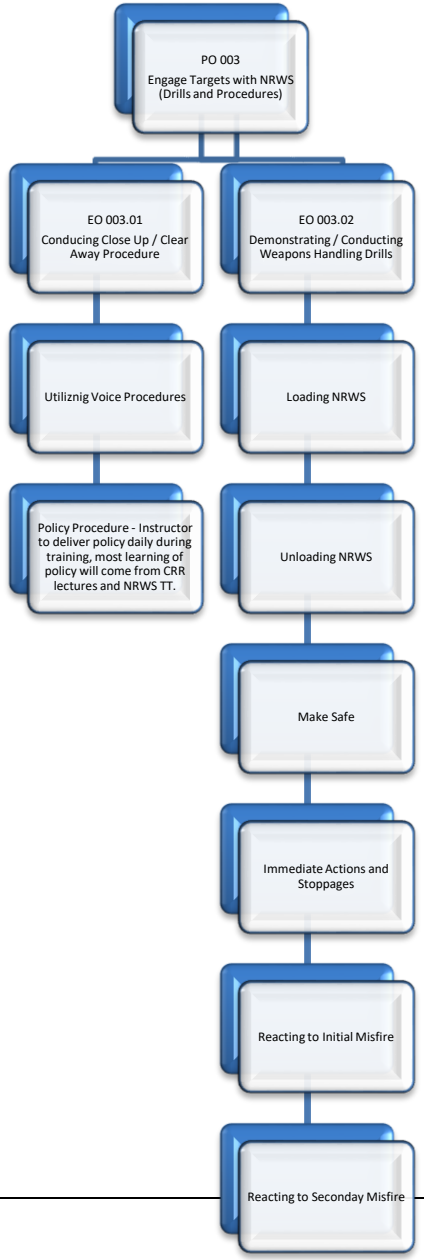
ANNEX D – TASK ANALYSIS DIAGRAM

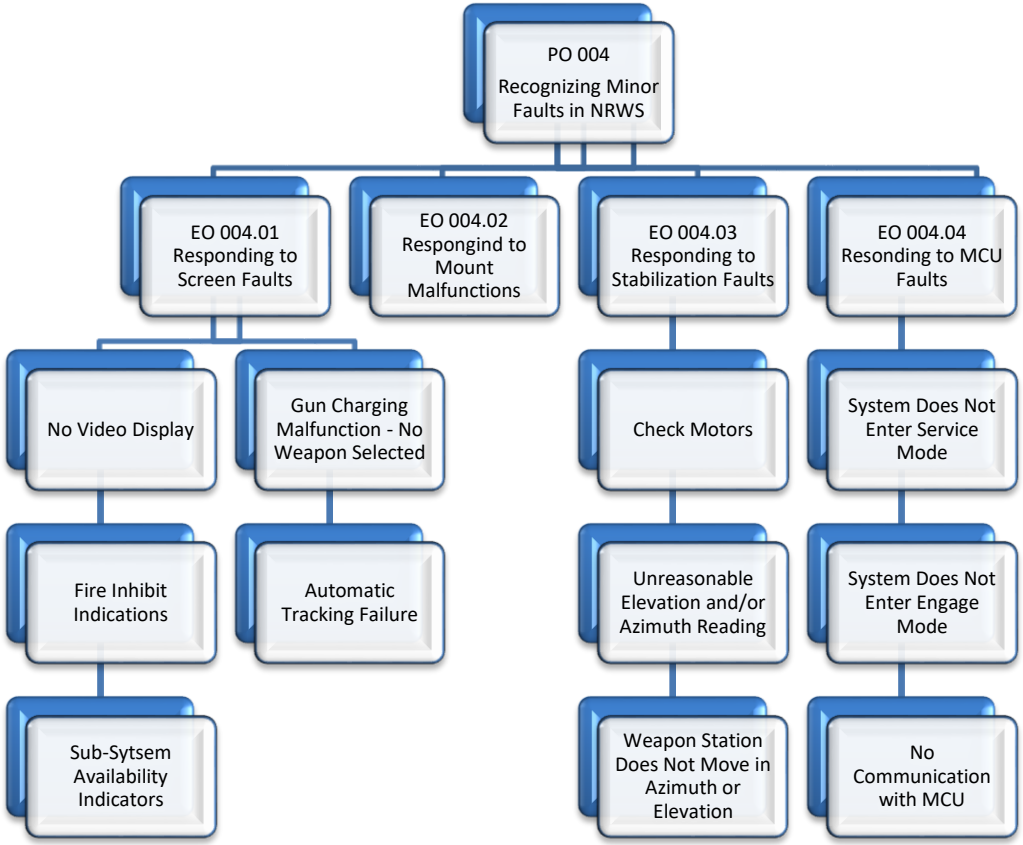
This diagram represents the original task analysis (scalar) used to define the job.




PO 001
Enable System For Weapon Use
EO 001.01
Mount 50 Cal HMG / C6 GPMG
EO 001.01
Converting NRWS Mount from Manual to Remote
Identifying Major Components
Placing Retaining Pins in Correct Location (Based on Weapon)
Placing Weapon on Mount
Securing Retaining Pins
Ensuring MO set to Safe
Disengaging Locking Pins
Engagin Mount Servo's
Disengaging BLR and Weapon Safety


PO 002
Preparing the NRWS Remote Operating Console (ROC) for use
EO 002.01
Performing System Set Up
EO 002.02
Making System Mechanically and Electrically Safe
EO 002.03
Employing Modes of Operation
EO 002.04
Converting NRWS ROC from Remote to Manual
Identifying Major Components
Turning On System
Performing Built In Test (BIT)
Conducting Communications Checks
Verifying Cross and Slave Functions
Operating EOSS In Surveillance Mode
Operating EOSS In Engagement Mode
Verifying Weapon and Ammunition Settings
Employing NRWS in Surveillance Mode
Employing NRWS in Engagement Mode
Employing NRWS in Offset Mode
Employing NRWS in Home Mode
Employing NRWS in Service Mode
Setting MCP to Main
Selecting Home Mode of Operation
Making NRWS Mechanically Safe


PO 003
Engage Targets with NRWS (Drills and Procedures)
EO 003.01
Conducing Close Up / Clear Away Procedure
EO 003.02
Demonstrating / Conducting Weapons Handling Drills
Utiliznig Voice Procedures
Loading NRWS
Policy Procedure - Instructor to deliver policy daily during training, most learning of policy will come from CRR lectures and NRWS TT.
Unloading NRWS
Make Safe
Immediate Actions and Stoppages
Reacting to Initial Misfire
Reacting to Seconday Misfire


PO 004
Recognizing Minor Faults in NRWS
EO 004.01 Responding to Screen Faults
EO 004.02 Respongind to Mount Malfunctions
EO 004.03 Responding to Stabilization Faults
EO 004.04 Resonding to MCU Faults
No Video Display
Gun Charging Malfunction - No Weapon Selected
Check Motors
System Does Not Enter Service Mode
Fire Inhibit Indications
Automatic Tracking Failure
Unreasonable Elevation and/or Azimuth Reading
System Does Not Enter Engage Mode
Sub-Sytsem Availability Indicators
Weapon Station Does Not Move in Azimuth or Elevation
No Communication with MCU

ANNEX E – COSTING INFORMATION

This table provides an estimated cost to deliver this training. Considered costs include Temporary Duty (TD), material, and personnel costs. TD costs include Rations & Quarters (R&Q) costs for relevant TE, and National Joint Counsel factors (meals & incidentals). Material costs are best estimates. Personnel costs are IAW the Cost Factors Manual for the current fiscal year for all instructors identified in Chapter 1 - Required Course Staffing.

TD Costs	# Trainees + Instructors /	Unit Price	# Days	Initial Course	Each Successive Course	5 Year Total
Airfare (Return)	1	$1000.00	1	$1000.00	$1000.00	$5000.00
JR R&Q	1	$48.64	1	$48.64	$1167.36	$5836.80
SR R&Q	1	$80.14	1	$80.14	$1923.36	$9616.80
Incidentals	1	$17.30	1	$17.30	$415.20	$2076.00
Subtotal				**$1146.08**	**$3505.92**	**$137529.60**
Material Costs	**Quantity**	**Unit Price**	**# Days**	**Initial Course**	**Each Successive Course**	**5 Year Total**
PP&S (Estimate)	16	$50.00	1	$800.00	$800.00	$4000.00
Classroom (square meters)	30	$13.00	1	$13.00	$0	$13.00
Computer requirements	16	$1000.00	1	$16000.00	$0	$16000.00
IT Equip (Smart board, etc.)	1	$6000.00	1	$6000.00	$0	$6000.00
DWAN connection (Setup)	1	$5000.00	1	$5000.00	$0	$5000.00
Link Toolkit (on DVD)	15	$1.00	1	$15	$15	$75
Instructional Materials (Printed / Handouts etc.)	16	$30.00	1	$480	$480	$2400.00
Subtotal				**$28308.00**	**$1295.00**	**$33488.00**
Weapon Material Costs	**Quantity**	**Unit Price**	**# Days**	**Initial Course**	**Each Successive Course**	**5 Year Total**
Computer Based Trainer	16	$105000.00	1	$105000.00	$0	$105000.00
NRWS Station	1	$450000.00	1	$450000.00	$0	$450000.00
50 Cal HMG	1	$12000.00	1	$12000.00	$0	$12000.00

Cleaner, Lubricant	1	$3.60	1	$3.60	$3.60	$18.00
Swabs	1	$4.17	1	$4.17	$4.17	$20.85
C6 GPMG	1	$7494.12	1	$7494.12	$0	$7494.12
7.62mm Dummy Ammo	15	$2.06	1	$30.90	$0	$30.90
50 Cal Dummy Ammo	15	$3.45	1	$51.75	$0	$51.75
Hand Linker/De-linker	1	$250.00	1	$250.00	$0	$250.00
Subtotal				**$574834.54**	**$7.77**	**$574865.62**
Total Estimated Costs				**$604288.62**	**$4808.69**	**$745883.22**

ANNEX F – ASSESSMENT PLAN

This table summarizes assessment (EC & PC) details contained within this QSP.

EC/PC	Scope	Purpose	How	Pass/Fail Criteria
EC 001	EO 001	To assess trainee's competence in taking and passing control of the NRWS Mount.	Practical	Trainee must obtain 19 / 27 SATs IAW EC 001 checklist.
PC 001	PO 001	To assess trainee's competence in taking and passing control of the NRWS Mount.	Practical	Trainee must obtain 23 / 32 SATs IAW PC 001 checklist. Impact to Failure: Trainee will be given counselling with practice, a supplemental PC, and following additional failure TRB process will be initiated IAW NPTORD 4500-54
EC 002	EO 002	To assess trainee's competence in knowledge and ability to employ the various modes of the NRWS.	Practical	Trainee must obtain 8 / 11 SATs IAW EC 002 checklist.
PC 002	PO 002	To assess trainee's competence in preparing the NRWS Remote Operating Console for Use.	Practical	Trainee must obtain 13 / 18 SATs IAW PC 002 checklist. Impact to Failure: Trainee will be given counselling with practice, a supplemental PC, and following additional failure

EC/PC	Scope	Purpose	How	Pass/Fail Criteria
				TRB process will be initiated IAW NPTORD 4500-54
EC 003	EO 003	To assess trainee's competence in making the system mechanically and electrically safe, conducting close up and clear away procedures, and conducting weapon handling drills.	Practical	Trainee must obtain 20/28 SATs on Part A, 19/26 SATs on Part B IAW EC 003 checklist.
PC 003	PO 003	To assess trainee's competence in making the system mechanically and electrically safe, conducting close up and clear away procedures, and conducting weapon handling drills.	Practical	Trainee must obtain 20/28 SATs on Part A, 19/26 SATs on Part B IAW PC 003 checklist. Impact to Failure: Trainee will be given counselling with practice, a supplemental PC, and following additional failure TRB process will be initiated IAW NPTORD 4500-54
EC 004	EO 004	To assess trainee's competence in responding to stabilization faults to include unreasonable elevation and/or azimuth readings, weapon station does not move in azimuth or elevation, system not entering engage mode, and no communication with MCU.	Practical	Trainee must obtain 4 / 5 SATs IAW EC 004 checklist.

EC/PC	Scope	Purpose	How	Pass/Fail Criteria
PC 004	PO 004	To assess trainee's competence in correcting minor faults to include Fire inhibit indications, sub-system availability indicators, no video display, and gun charging malfunction.	Practical	Trainee must obtain 3 / 4 SATs IAW PC 004 checklist. Impact to Failure: Trainee will be given counselling with practice, a supplemental PC, and following additional failure TRB process will be initiated IAW NPTORD 4500-54

ANNEX G – TEL DEVELOPMENT OPPORTUNITIES

If a training methodology is determined to be ineffective and/or inefficient, a Learning Support Request (LSR) may be submitted to correct the deficiency. Technology Enabled Learning (TEL) development opportunities are assessed IAW NPTORD 4500-84 following submission.

Title	Requirement Description	Lesson #	LSR #
NRWS Mount Card Sort Activity	Poster of EO 002.01 with card sort	EO 002.01	2022-40-DECK
NRWS Submenu Layout Poster	Poster of NRWS ROC Submenu Layout	EO 002.01	52-21
Policy Order Printout	Print out of the Policy Orders for each CBT and instructor station.	PO 003	2022-44-DECK

ANNEX H – SCHEDULING INFORMATION

This table summarizes the conditions cell in each LG. The order within the table reflects the order of instruction. All timings are based on maximum load. This is required for the Scheduling Cell and completion is mandatory.

Lesson		Classroom				IT Assets (Trainees & Instructors)					
#	Time (minutes)	Location	Classification	White Board	Desks	DWAN	CSNI	Secret Stand Alone	GP Net	Projector or TV	TEL & Other
PO 001 – INDENTIFYING MAJOR COMPONENTS											
EO 001.01	200	NRWS Classroom	UNCLASS	1	8	0	0	0	0	0	0
EO 001.02	100	NRWS Classroom	UNCLASS	1	8	0	0	0	0	0	0
EC 001	100	NRWS Classroom	UNCLASS	1	8	0	0	0	0	0	0
PC 001	150	NRWS Classroom	UNCLASS	1	8	0	0	0	0	0	0
PO 002 – OPERATE THE NRWS REMOTE OPERATING CONSOLE											
EO 002.01	100	NRWS Classroom	UNCLASS	1	8	0	0	0	0	0	0
EO 002.02	50	NRWS Classroom	UNCLASS	1	8	1	0	0	0	1	Projector
EO 002.03	50	NRWS Classroom	UNCLASS	1	8	0	0	0	0	0	0
EO 002.04	50	NRWS Classroom	UNCLASS	1	8	0	0	0	0	0	0
EC 002	100	NRWS Classroom	UNCLASS	1	8	0	0	0	0	0	0
PC 002	150	NRWS Classroom	UNCLASS	1	8	0	0	0	0	0	0
PO 003 – OPERATE NAVAL REMOTE WEAPON SYSTEM											
EO 003.01	200	NRWS Classroom	UNCLASS	1	8	0	0	0	0	0	0
EO 003.02	600	NRWS Classroom	UNCLASS	1	8	0	0	0	0	0	0

Lesson		Classroom				IT Assets (Trainees & Instructors)					
#	Time (minutes)	Location	Classification	White Board	Desks	DWAN	CSNI	Secret Stand Alone	GP Net	Projector or TV	TEL & Other
EC 003	250	NRWS Classroom	UNCLASS	1	8	0	0	0	0	0	0
PC 003	250	NRWS Classroom	UNCLASS	1	8	0	0	0	0	0	0
PO 004 – CORRECT MINOR NRWS FAULTS											
EO 004.01	50	NRWS Classroom	UNCLASS	1	8	0	0	0	0	1	0
EO 004.02	50	NRWS Classroom	UNCLASS	1	8	0	0	0	0	1	0
EO 004.03	50	NRWS Classroom	UNCLASS	1	8	0	0	0	0	1	0
EC 004	50	NRWS Classroom	UNCLASS	1	8	0	0	0	0	1	0
PC 004	50	NRWS Classroom	UNCLASS	1	8	0	0	0	0	1	0